

or exosuit. It basically consists of an outer framework which is put on like a dress, and a powered system of motors or hydraulics contribute some or complete energy for limb movement. Since it very much resembles the external skeleton that supports and protects bodies of insects and other animals, it has earned the name exoskeleton.

The main purpose of a powered exoskeleton is to impart more strength and fortitude to the wearer. These machines are mostly used in Defence to aid soldiers lug heavy items and sometimes more than that. The US Army is in process of creating an Iron Man-like suit that imparts superhuman strength to the user. This exoskeleton is called the Tactical Assault Light Operator Suit (TALOS) and will have strata of smart materials built-in with sensors as well as a wearable computer very much like Google Glass. It can keep an eye on soldiers' critical parameters and multiply the available strength using hydraulics.

Powered exoskeletons also enable firemen and other rescue workers endure hazardous conditions. In the healthcare sector their application ranges from carrying out very accurate surgeries to enabling nurses to shift bulky patients. And then there are several other very important uses for people with disabilities and special needs. ReWalk is a commercial exoskeleton and a bionic walking assistance device that uses powered leg attachments to enable paraplegics to stand straight, walk and climb steps.

The system is power-driven by a backpack battery and controlled by a simple wrist-mounted remote which detects and enhances the user's movement. There are several challenges that need to be overcome before powered exoskeletons can become popular. One of the most potent issues is creation of a high yet compact power supply that is large enough to enable an exoskeleton to function for long periods without any external power source. Right now multiple industries have pooled in their expertise and are working in close collaboration with R&D

organisations, government labs as well as academia to overcome the hurdles.

'Wear' do we go from here

Today, we use corneal transplants to mend defective eyes. In the future, we may simply be able to insert artificial eyes with zoom capabilities, infrared sensors and night vision. We already have The MindMesh, an EEG (electroencephalogram)-based 'thinking cap' that allows the user to hook different gadgets to their brain.

For example, a blind person can plug in a camera and use it as an 'eye.' In time to come, wearable will evolve into 'embeddable,' which could be sensors placed under the skin or swallowed. Today, the Proteus smart pill is capable of communicating a text to a doctor or some well-wisher from within the body to convey that the medicine has been consumed. Some years from now, we are likely to have reached a stage wherein ingestible devices will be transmitting images from inside the human gastrointestinal tract to help doctors diagnose the cause and bring down the need for surgery and other invasive procedures.

The market potential for wearable computing stems from its diversity—from watches and fitness trackers to sensor-filled clothes and Google Glass. In December 2013, Google Glass informed us via its Google+ social network page that users of its Internet-linked eyewear will now be able to click photos merely by winking. Now there are speculations that soon you and I will be able to pay the taxi driver simply by winking at the cab meter, and what not. As per Credit Suisse, a leading financial services company, the market for wearable technology will rise tenfold to as high as 50 billion US dollars by 2016 or 2018. The next big revolution in information technology will not reside in your pocket or on your office table; you will be wearing it! ●

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